

Solution 02
Coordinate Transform and Lyapunov Stability

Exercise 1

Consider the following linear discrete-time system

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k), \\ y(k) &= Cx(k) + Du(k),\end{aligned}$$

and assume A allows an eigendecomposition¹ $A = W\Delta W^{-1}$, where $W = [w_1, \dots, w_n]$ is the matrix of eigenvectors w_i , and $\Delta = \text{diag}([\lambda_1, \dots, \lambda_n])$ is the corresponding vector of eigenvalues λ_i .

- Find a coordinate transformation $\tilde{x}(k) = Tx(k)$ such that $\tilde{x}(k+1) = \tilde{A}\tilde{x}(k) + \tilde{B}u(k)$ with diagonal $\tilde{A}$. Find the expressions for $\tilde{A}$, $\tilde{B}$, $\tilde{C}$ and $\tilde{D}$.

Consider now the autonomous system, i.e., $u(k) = 0$ for all $k \geq 0$, and assume that the absolute value of all eigenvalues of A are $|\lambda_i| < 1$.

- Show that the infinite horizon cost $\sum_{k=0}^{\infty} x(k)^T Q x(k)$ is bounded for all $Q > 0$ (Q positive definite).

Hints:

- A symmetric real matrix Q is positive definite if $z^T Q z > 0$ for every column vector $z \neq 0$
- Every matrix $Q > 0$ can be decomposed as $Q = Q^{\frac{1}{2}} Q^{\frac{1}{2}}$ with $Q^{\frac{1}{2}} > 0$
- $\|Ax\|_2 \leq \|A\|_2 \|x\|_2 = \sqrt{\lambda_{\max}(A^T A)} \|x\|_2$, where $\lambda_{\max}(A^T A)$ is the maximum eigenvalue of $A^T A$
- $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$ for $|r| < 1$

- Show that for every $Q > 0$ there exists $P > 0$ satisfying the Lyapunov equation

$$\underbrace{x(k+1)^T P x(k+1)}_{V(x(k+1))} - \underbrace{x(k)^T P x(k)}_{V(x(k))} = - \underbrace{x(k)^T Q x(k)}_{\alpha(x(k))}.$$

Solution:

- (Lec. 1c, Slide 32 - 33) Given an invertible matrix T , it holds that

$$\begin{aligned}\tilde{x}(k) &= Tx(k) \\ \Leftrightarrow x(k) &= T^{-1}\tilde{x}(k).\end{aligned}$$

Using this transformation results in the following system equation

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\ \Leftrightarrow T^{-1}\tilde{x}(k+1) &= AT^{-1}\tilde{x}(k) + Bu(k) \\ \Leftrightarrow \tilde{x}(k+1) &= \underbrace{TAT^{-1}}_{\tilde{A}}\tilde{x}(k) + \underbrace{TB}_{\tilde{B}}u(k)\end{aligned}$$

and

$$y(k) = \underbrace{CT^{-1}}_{\tilde{C}}\tilde{x}(k) + \underbrace{D}_{\tilde{D}}u(k).$$

¹i.e. $A \in \mathbb{R}^{n \times n}$ has n linearly independent eigenvectors. The statements in 2) and 3) hold regardless of this assumption.

Taking the eigenvalue decomposition of

$$A = W\Delta W^{-1}$$

$$\Leftrightarrow \underbrace{W^{-1}AW}_T = \underbrace{\Delta}_{\tilde{A} \text{ diagonal}}$$

the resulting matrices of the transformed system are given by

$$\begin{aligned}\tilde{A} &= \Delta, \\ \tilde{B} &= W^{-1}B, \\ \tilde{C} &= CW, \\ \tilde{D} &= D.\end{aligned}$$

Note:

- The transformation does not change the behaviour from $u(k)$ to $y(k)$
- The transformation can simplify the system analysis: $\tilde{x}_1(k+1) = \lambda_1 \tilde{x}_1(k) \rightarrow$ stable if $|\lambda_1| < 1$
- The transformation is only possible if A has eigenvalue decomposition

2. Using the coordinate transform and the given hints, rewrite the infinite horizon cost as follows

$$\begin{aligned}\sum_{k=0}^{\infty} x(k)^T Q x(k) &= \sum_{k=0}^{\infty} \tilde{x}(k)^T \underbrace{(T^{-1})^T Q T^{-1}}_{\tilde{Q} > 0} \tilde{x}(k) \\ &= \sum_{k=0}^{\infty} \tilde{x}(k)^T \tilde{Q} \tilde{x}(k) \\ &= \sum_{k=0}^{\infty} \tilde{x}(k)^T \tilde{Q}^{\frac{1}{2}} \tilde{Q}^{\frac{1}{2}} \tilde{x}(k) \\ &= \sum_{k=0}^{\infty} \|\tilde{Q}^{\frac{1}{2}} \tilde{x}(k)\|_2^2 \\ &\leq \sum_{k=0}^{\infty} \left(\|\tilde{Q}^{\frac{1}{2}}\|_2 \|\tilde{x}(k)\|_2 \right)^2 \\ &= \underbrace{\|\tilde{Q}^{\frac{1}{2}}\|_2^2}_{=\lambda_{\max}(\tilde{Q})} \sum_{k=0}^{\infty} \|\tilde{x}(k)\|_2^2.\end{aligned}$$

To bound the infinite horizon cost, we now need to bound $\sum_{k=0}^{\infty} \|\tilde{x}(k)\|_2^2$:

$$\begin{aligned}\|\tilde{x}(k+1)\|_2 &= \left\| \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}}_{\tilde{A}} \tilde{x}(k) \right\|_2 \leq |\lambda_i(A)|_{\max} \|\tilde{x}(k)\|_2 \\ \rightarrow \|\tilde{x}(k)\|_2 &\leq |\lambda_i(A)|_{\max}^k \|\tilde{x}(0)\|_2,\end{aligned}$$

where $|\lambda_i(A)|_{\max}$ is the maximum absolute value of all eigenvalues of A . This results in the sum

$$\sum_{k=0}^{\infty} \|\tilde{x}(k)\|_2^2 \leq \|\tilde{x}(0)\|_2^2 \sum_{k=0}^{\infty} |\lambda_i(A)|_{\max}^{2k} = \frac{\|\tilde{x}(0)\|_2^2}{1 - |\lambda_i(A)|_{\max}^2},$$

since $|\lambda_{\max}(A)| < 1$. We finally have

$$\sum_{k=0}^{\infty} x(k)^T Q x(k) \leq \lambda_{\max}(\tilde{Q}) \frac{\|\tilde{x}(0)\|_2^2}{1 - |\lambda_i(A)|_{\max}^2}.$$

3. The infinite horizon cost at time step 0 can be expressed as a function of the state $x(0)$ and matrix P (Lec. 1c, Slide 56) through

$$\sum_{k=0}^{\infty} x(k)^T Q x(k) = x(0)^T \underbrace{\left(\sum_{k=0}^{\infty} (A^k)^T Q A^k \right)}_{:=P>0} x(0),$$

since $x(k) = A^k x(0)$. Inserting the above expression for P into the quadratic Lyapunov function $V(x(k)) = x(k)^T P x(k)$ leads to

$$\begin{aligned} x(k)^T P x(k) &= x(k)^T \left(\sum_{i=0}^{\infty} (A^i)^T Q A^i \right) x(k) \\ &= x(k)^T Q x(k) + x(k)^T \left(\sum_{i=1}^{\infty} (A^i)^T Q A^i \right) x(k) \\ &= x(k)^T Q x(k) + x(k)^T A^T \left(\sum_{i=1}^{\infty} (A^{i-1})^T Q A^{i-1} \right) A x(k) \\ &= x(k)^T Q x(k) + x(k)^T A^T \left(\sum_{j=0}^{\infty} (A^j)^T Q A^j \right) A x(k) \\ &= x(k)^T Q x(k) + x(k+1)^T P x(k+1), \end{aligned}$$

with $j = i - 1$. Reordering this equation results in the Lyapunov equation

$$\underbrace{x(k+1)^T P x(k+1)}_{V(x(k+1))} - \underbrace{x(k)^T P x(k)}_{V(x(k))} = - \underbrace{x(k)^T Q x(k)}_{\alpha(x(k))},$$

i.e., the infinite horizon cost can be determined by solution P of the Lyapunov equation.

Exercise 2

This exercise investigates stability properties of the origin for the following nonlinear discrete-time system

$$x(k+1) = \underbrace{\frac{x(k) + \sin(x(k))}{4}}_{:=f(x)}. \quad (1)$$

1. What can be concluded about stability of the equilibrium point $x_s = 0$ for system (1) using Lyapunov's **indirect** method?
2. What can be concluded about stability of the equilibrium point $x_s = 0$ for system (1) using Lyapunov's **direct** method?

Solution:

1. For Lyapunov's indirect method, we investigate the linearized system $x(k+1) = A x(k)$ with

$$A = \left. \frac{\partial f(x)}{\partial x} \right|_{x=0} = \left. \frac{1 + \cos(x)}{4} \right|_{x=0} = \frac{1}{2}. \quad (2)$$

For the eigenvalue λ of A it holds $|\lambda| = |1/2| < 1$ and we can therefore conclude that the origin is locally asymptotically stable for system (1). As the linearization used in Lyapunov's indirect method is only valid close to the origin, we cannot conclude global stability.

2. For Lyapunov's direct method we need to find a Lyapunov function $V(x)$ which satisfies the following conditions (Slide 49 Lec.1c):

- $\|x\| \rightarrow \infty \Rightarrow V(x) \rightarrow \infty$.

- $V(0) = 0$ and $V(x) > 0, \forall x \in \mathbb{R} \setminus \{0\}$.
- $V(f(x)) - V(x) \leq -\alpha(x), \forall x \in \mathbb{R}$ where $\alpha(x)$ has to be continuous positive definite.

We choose the Lyapunov function $V(x) = x^2$ and see that the first and second condition are trivially verified. For the second condition, we use $|\sin(x)| \leq |x|$ to derive

$$\begin{aligned}
 V(f(x)) - V(x) &= \left(\frac{x + \sin(x)}{4} \right)^2 - x^2 \\
 &= \frac{(x^2 + 2x\sin(x) + \sin(x)^2)}{16} - x^2 \\
 &\leq \frac{4}{16}x^2 - x^2 \\
 &= - \underbrace{\frac{3}{4}x^2}_{:=\alpha(x)>0}.
 \end{aligned} \tag{3}$$

As system (1) admits a valid Lyapunov function $V(x)$, we can conclude the origin $x = 0$ is in fact globally asymptotically stable.